

A Power BI Based Integrated Sales Intelligence & Performance Analysis Dashboard

PROJECT DOCUMENTATION

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Course: AI-driven Data Analytics

Tool Used: Microsoft - Power BI

PROJECT OVERVIEW

This project presents an interactive sales performance dashboard developed in Power BI, utilizing multiple datasets including orders, order details, and monthly sales targets. The analysis covers three primary categories - Furniture, Clothing, and Electronics. Key objectives include leveraging DAX to create calculated measures such as Total Sales, Total Profit, Profit Margin, and Target Variance, and performing a comparative analysis of actual performance against predefined targets. The dashboard provides insights into category-wise performance, regional trends, and sub-category profitability to support data-driven decision-making.

OBJECTIVES OF THE PROJECT

The primary objective of this project is to design and develop a comprehensive Sales Performance Dashboard using Microsoft Power BI.

The project aims to:

1. **Integrate and Model Data:** To integrate data from multiple sources - List of Orders, Order Details, and Sales Targets - and establish relationships through data cleaning and data modeling.
2. **Develop KPIs using DAX:** To create key performance measures using DAX, such as Total Sales, Total Profit, Profit Margin, Target Variance, and Achievement Percentage.
3. **Actual vs. Target Analysis:** To compare actual sales performance against predefined monthly sales targets across different categories, segments, and regions.
4. **Identify Trends and Patterns:** To analyze sales trends across categories (Furniture, Clothing, Electronics), sub-categories, states, and cities to identify top-performing and underperforming areas.
5. **Enable Data-Driven Decision Making:** To provide an interactive and visually appealing dashboard that enables stakeholders to make informed business decisions.

DATASETS USED

- **List of Orders:** This is the primary fact table containing order-level information. It includes fields such as Order ID, Order Date, Customer details, Segment (Consumer, Corporate, Home Office), State, City, and Category. It provides the high-level overview of each transaction.
- **Order Details:** This is the detailed transactional dataset which provides a granular breakdown of each order. It includes attributes like Sub-Category, Quantity Sold, Unit Price, Total Amount, Profit, and Discount. This table is crucial for calculating core KPIs like Total Sales and Profit Margin.
- **Sales Target:** This is the benchmark dataset containing the predefined monthly sales targets. It includes fields such as Month, Category, and Target Amount. This dataset is used to compare actual performance against expected goals and to calculate Variance and Achievement %.

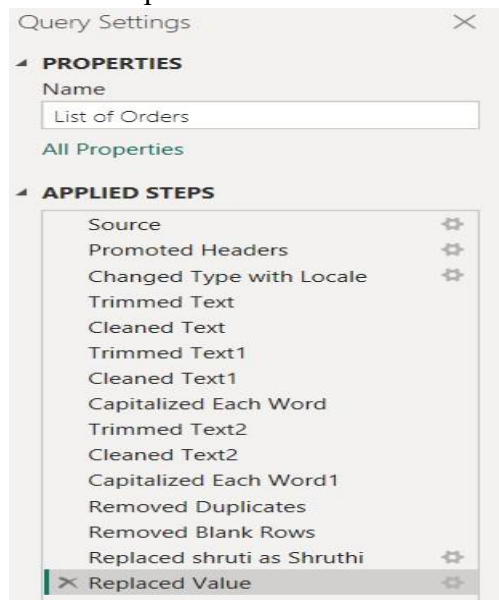
DATA PREPARATION AND DATA CLEANING

The data preparation phase involved importing three raw datasets - List of Orders, Order Details, and Sales Target - into Power BI Power Query Editor. Extensive cleaning and transformation were performed to ensure data consistency, accuracy, and readiness for modeling.

Key transformation steps included:

1. **Duplicate Removal:** Verified data integrity by identifying and removing duplicate Order IDs from the List of Orders table to ensure each transaction was counted only once.
2. **Data Cleaning and Text Standardization:** Applied TRIM and CLEAN functions in Power Query to eliminate leading/trailing spaces and non-printable characters from key textual columns such as Customer Name, State, and City, ensuring uniformity for accurate grouping and mapping.
3. **Data Type Correction:** Standardized data types across all tables - converted Order Date and Target Month to Date format, and Sales, Profit, and Quantity to numeric format to enable accurate DAX calculations.
4. **Handling Null and Missing Values:** Identified and handled null values in critical columns like State and Sub-Category by replacing them with "Unknown" or filtering them out based on business logic to avoid calculation errors.
5. **Creation of Calculated Columns:** Created new columns such as Full Location (City + State + Country) for accurate geographic mapping, and Month-Year for time-series analysis.

After these transformations, the clean datasets were loaded into the data model for establishing relationships.



DATA MODELING AND RELATIONSHIPS

After data cleaning and transformation, a relational data model was established in Power BI's Model View to ensure seamless data flow and accurate cross-table calculations. A Star Schema approach was adopted, where dimension tables are linked to central fact tables.

Relationships Established:

1. List of Orders & Order Details (One-to-Many):

A One-to-Many relationship was created using Order ID as the primary key. One Order in the List of Orders can have multiple line items in Order Details. This is the core relationship of the model and is set to active with single-directional filtering.

2. Sales Target & List of Orders (Many-to-Many / Logical Relationship):

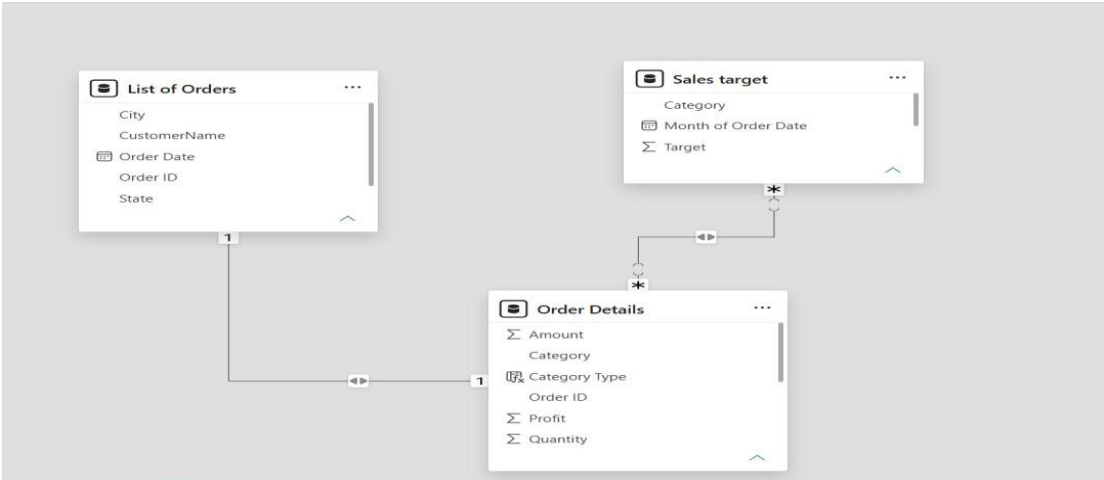
A relationship was established based on Category and Month-Year. Since Sales Target contains monthly targets per Category, it was linked to the Orders table to enable Actual vs. Target comparison. A bridge date table was created to support time-intelligence analysis.

3. Date Table (One-to-Many):

A dedicated Calendar Table was created using DAX CALENDAR() function and linked to both List of Orders (Order Date) and Sales Target (Target Month) to enable time-based analysis such as Year, Quarter, and Month-wise trends.

All relationships were configured with proper cardinality and cross-filter direction to avoid ambiguity and ensure that DAX measures produce accurate results when sliced by Category, State, or Month.

Model Type: Star Schema
Relationship Type: Active, Single Direction



CALCULATED COLUMNS

Calculated columns were created using DAX (Data Analysis Expressions) at the table level to enrich the existing datasets and enable more granular and dynamic analysis. Unlike measures, which are calculated on-the-fly, calculated columns are computed row-by-row and stored in the model.

Key Calculated Columns Created:
Category Type

1 Category Type = 'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]									
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue per Order	Sales Category	
B-25602	561	212	3	Clothing	Saree	Clothing - Saree	1683	Above Average	
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree	952	Below Average	
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree	579	Below Average	
B-25604	157	5	9	Clothing	Saree	Clothing - Saree	1413	Below Average	
B-25605	75	0	7	Clothing	Saree	Clothing - Saree	525	Below Average	
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree	100	Below Average	
B-25610	43	0	3	Clothing	Saree	Clothing - Saree	129	Below Average	
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree	320	Below Average	
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree	14427	Above Average	
B-25619	353	90	8	Clothing	Saree	Clothing - Saree	2824	Above Average	
B-25622	534	0	3	Clothing	Saree	Clothing - Saree	1602	Above Average	
B-25623	149	-87	4	Clothing	Saree	Clothing - Saree	596	Below Average	
B-25625	635	-349	5	Clothing	Saree	Clothing - Saree	3175	Above Average	

Revenue per Order

The Revenue per Order is a calculated column created in the Order Details table using DAX to determine the monetary value generated from each individual transaction. It calculates the total revenue by multiplying Quantity and Unit Price.

Category Type	Revenue per Order	Sales Category
Clothing - Saree	1683	Above Average
Clothing - Saree	952	Below Average
Clothing - Saree	579	Below Average
Clothing - Saree	1413	Below Average
Clothing - Saree	525	Below Average
Clothing - Saree	100	Below Average
Clothing - Saree	129	Below Average
Clothing - Saree	320	Below Average
Clothing - Saree	14427	Above Average
Clothing - Saree	2824	Above Average
Clothing - Saree	1602	Above Average

Sales Category

This analysis focuses on evaluating the sales performance across the three primary business categories - Furniture, Clothing, and Electronics - to understand their individual contribution to the overall revenue.

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```
1 Sales Category =
2 IF(
3     'Order Details'[Amount] > AVERAGE('Order Details'[Amount]),
4     "Above Average",
5     "Below Average"
6 )
```

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue per Order	Sales Category
B-25602	561	212	3	Clothing	Saree	Clothing - Saree	1683	Above Average
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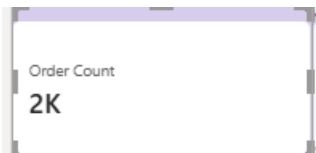
DAX MEASURES

DAX (Data Analysis Expressions) measures are dynamic calculations created in Power BI to perform aggregations across the data model.

Unlike calculated columns which are computed row-by-row, measures are calculated on-the-fly based on the filter context of the visuals, enabling highly interactive analysis.

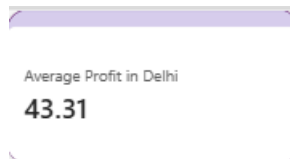
Order Count

Order Count = `COUNT('Order Details'[Order ID])`



Average Profit in Delhi

Average Profit in Delhi = `CALCULATE(AVERAGE('Order Details'[Profit]),'List of Orders'[City]="Delhi")`



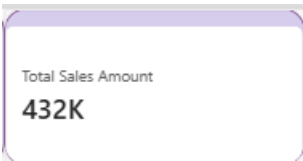
Year-to-Date (YTD) Sales

Year To Date Sales = `TOTALYTD(SUM('Order Details'[Amount]),'List of Orders'[Order Date])`



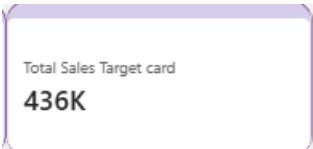
Total Sales

Total Sales = `SUM('Order Details'[Amount])`



Total Target

Total Target = `SUM('Sales target'[Target])`



VISUALIZATIONS CREATED



VISUALIZATION SCREENSHOTS AND EXPLANATIONS

KEY INSIGHTS & FINDINGS

The interactive Power BI dashboard, built by integrating List of Orders, Order Details, and Sales Target datasets, provides a comprehensive view of business performance.

The following data-driven insights were derived:

Category-wise Performance:

Furniture: Emerged as the highest revenue-generating category in terms of Total Sales. However, it consistently under-achieved its monthly sales targets, indicating a gap between expected and actual demand.

Clothing: Demonstrated the most stable and consistent growth. It frequently met and exceeded its sales targets, making it the most reliable category with strong customer demand.

Electronics: Showed highly volatile performance with significant peaks during promotional and festive months, suggesting its sales are highly driven by offers and discounts.

Profitability Analysis:

There is a notable difference between revenue and profitability. While Furniture leads in revenue, Clothing contributes the highest profit margin, making it the most profitable segment.

Sub-categories such as Printers and Bookcases show high sales volume but relatively low profit margins. This indicates a need for pricing strategy revision or cost optimization.

Regional & Geographical Insights:

State-wise: Maharashtra, Madhya Pradesh, and Rajasthan were identified as the top 3 key markets, contributing over 60% of total sales.

City-wise: Metropolitan cities like Mumbai, Indore, and Jaipur consistently ranked as top performers, highlighting the strong purchasing power in urban regions.

Customer & Seasonal Behavior

High-value orders and repeat purchases are concentrated in metropolitan regions, indicating stronger customer loyalty in urban markets.

A clear seasonal trend was observed with major sales spikes in April and December, correlating with financial year-end sales and festive season promotions.

CONCLUSION

This Power BI project successfully demonstrates the power of Business Intelligence in transforming raw, disparate datasets into strategic, actionable business insights.

By integrating three key datasets - List of Orders, Order Details, and Sales Target - into a unified Star Schema data model, a dynamic and interactive dashboard was developed. The dashboard provides a 360-degree view of business performance across multiple dimensions, including category, region, time, and profitability.

The project effectively bridges the gap between data and decision-making. Through the use of advanced DAX measures, calculated columns, and KPI-based visualizations, it enables stakeholders to not only track overall sales performance but also to perform in-depth Actual vs. Target analysis, identify high-performing markets, and pinpoint areas requiring strategic intervention.

In conclusion, the solution moves beyond simple reporting to deliver a data-driven decision support system that can aid in optimizing pricing strategies, improving regional targeting, and driving overall profitability and growth.